BENCHMARKING U-NET VARIANTS FOR LUNG SEGMENTATION IN CHEST X-RAYS

Akshat G[1], Divyansh Gupta[2], Rashmi R[3]

[1, 2, 3] *Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal, India.*

Corresponding author: r.rashmi@manipal.edu

Abstract. Automated Segmentation of Lung lobes in Chest X-ray images is a crucial and fundamental step in many clinical workflows. Most of the present deep learning methods face challenges with generalization due to limited datasets and training data. To overcome this, in this study, a new dataset was compiled. It consists of two publicly available, government-approved chest X-ray datasets. This combined dataset was used to train and evaluate four major U-Net architectures being, U-Net, ResU-Net, Attention U-Net and U-Net++. The results from this study demonstrate that these advanced U-Net variants can achieve high segmentation performance, with U-Net++ outperforming its predecessors in terms of Dice Coefficient and Jaccard Index. This study provides a robust solution for medical image segmentation and offers valuable insights into selecting optimal deep learning models for medical image segmentation with accessible datasets.

Keywords: Computer Aided Diagnosis, Medical Image Processing, Lung Segmentation, Deep Learning, U-Net

1 INTRODUCTION

Medical image analysis is crucial for supporting healthcare professionals in making timely diagnoses and treatment decisions. Currently, the analysis of medical images is performed by radiologists and other medical imaging specialists. While radiologists are essential, the process of medical image analysis can be time-consuming, and coupled with the high workload experienced by such professionals, the diagnosis is prone to observer variability and can lead to inconsistent diagnosis and treatment plans [1].

X-rays are used because they offer a cheap, fast, and non-invasive way to visualize the structures of the body. Lung X-rays are important for monitoring pulmonary diseases such as pneumonia, lung cancer, and tuberculosis. They also play an important role in monitoring disease progression in the body, such as tracking the progression of diseases like COVID-19 and assisting in evaluating treatment effectiveness. These X-rays are also used as a screening tool and for post-surgical procedures and ICU evaluation during trauma cases.

Automated lung segmentation can be used to isolate the region of interest and can improve the accuracy and efficiency of radiologists [2]. However, precise lung segmentation can be challenging due to the wide range of patient anatomy and anatomical noise (e.g., heart, ribs, clavicle, tissues). Sometimes lung boundaries can be fuzzy and blend with the surrounding tissues.

Although numerous U-Net variants have been proposed in recent years, a comprehensive analysis of their performance in tasks like lung segmentation is still lacking. The present work aims to bridge the gap by evaluating and comparing the performance of the four models as stated in **Section 3.3** under consistent conditions.

2 LITERATURE REVIEW

Traditional methods such as segmentation based on edge detection, thresholding, region growing, and clustering have been used for a long time, but struggle with noise sensitivity, weak boundaries and are unable to handle complex patterns from the medical images [3]. Deep Learning (DL) methods namely Convolutional Neural Networks (CNNs), enable automatic feature extraction directly from raw input image. This shift has significantly improved performance for tasks such as disease detection, localization, and segmentation [4][25].

U-Net has become one of the popular image segmentation architectures for segmenting region of interest from the medical images [5][25]. It gained popularity due to its ability to perform well even on small datasets and its effectiveness in segmenting medical structures with varying shapes and sizes. U-Net's design enables it to retain important spatial information and produce accurate segmentation masks, which made it a foundational model for many follow-up studies.

Over time, several variants have been proposed to address specific limitations of the original U-Net. ResU-Net, for instance, was introduced to improve training stability and feature learning, particularly in deeper networks [6].

Attention U-Net builds on the base architecture by selectively focusing on the most relevant parts of an image. This approach helps the model deal with cluttered or noisy anatomical structures, leading to more precise segmentation, particularly useful in real clinical scans where overlapping tissues and variable contrast are common challenges [7].

U-Net++ enhances the original model by reducing the semantic mismatch between the encoding and decoding paths. It introduces a more refined way of transferring information between network levels, resulting in better localization and boundary precision, especially in difficult cases where the lung edges are not clearly defined [8].

While each of these U-Net variants has been individually explored across different medical image segmentation tasks [25], prior literature typically evaluates them on separate datasets, with varying preprocessing steps, architectures, loss functions, and training schedules. This lack of standardization makes it difficult to draw consistent conclusions about their relative performance [3][4]. There is limited work that benchmarks these U-Net variants under unified, controlled settings on diverse chest X-ray datasets. Additionally, many evaluations focus exclusively on either healthy lungs or a single disease class, whereas real-world diagnostic settings often involve a wide range of pathological and anatomical variation [4][5].

Considering this, the present study aims to provide a structured comparison of the four models as stated in **Section 3.3** for binary lung segmentation, using consistent training conditions and evaluation metrics across a diverse dataset comprising both tuberculosis-affected and healthy chest X-rays.

3 METHODOLOGY

3.1 Dataset

The present study uses X-ray images from two publicly available sources:

- **Montgomery County (MC) chest X-ray set:** The dataset has a total of 138 chest X-rays, of which 59 are affected with manifestations of tuberculosis [9].
- **Shenzhen Dataset - Chest X-ray set:** The dataset contains 662 chest X-rays, of which 336 are affected with manifestations of tuberculosis [10].

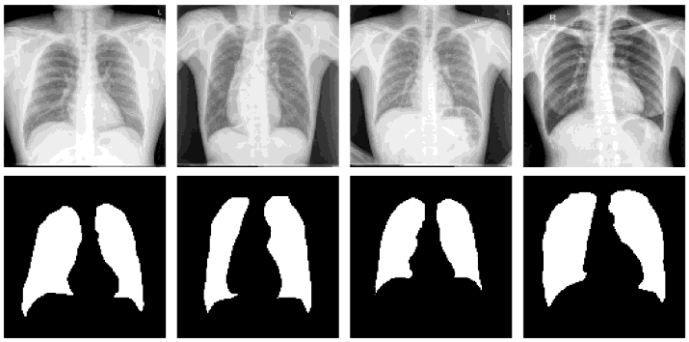

Fig. 1. Lung X-Ray sample images from the dataset.

In **Fig. 1.**, the top row shows the original grayscale Lung X-ray from both the above-mentioned datasets. The bottom row presents the corresponding ground truth binary masks.

3.2 Preprocessing

Since both the datasets are individually small to train a robust model, the datasets were pre-processed and combined using NumPy libraries. This combined dataset comprises a total of 704 images consisting of both healthy and TB-affected lung X-rays. All images were resized using bilinear interpolation to a consistent resolution of 256x256 pixels. The images were normalized using min-max normalization technique and were converted into a Keras compatible array. The images were later shuffled thoroughly to ensure the images from the two datasets are randomly present throughout the dataset. In this present study, both the datasets have been combined and split into two sets; training and testing, which ensures that the model was evaluated on unseen data to aid in quantitative analysis of the generalizability of the model.

3.3 Model Architecture

This study evaluates four prominent U-Net models (U-Net, ResU-Net, Attention U-Net and U-Net++) for lung segmentation to draw careful differences between each of them. These models were selected due to their proven effectiveness in medical image segmentation [5][6][7][8].

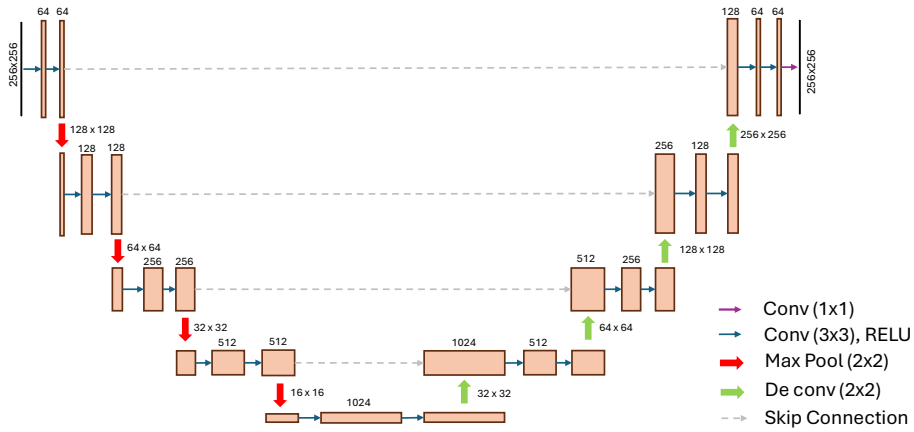

Fig. 2. U-Net architecture.

Fig. 2. shows the U-Net architecture that was employed for this study. The model is inspired by original U-Net model [5], which uses standard encoders and decoders for down-sampling and up-sampling respectively.

Attention U-Net model follows a similar architecture as U-Net but incorporates attention-gates [7]. These gates help aid the model to selectively focus on the relevant segmentation features. The attention-gates are added in between skip connections and de-convolution layers. The architecture of the attention-gates used in this study to train the model has been shown in **Fig. 3.**

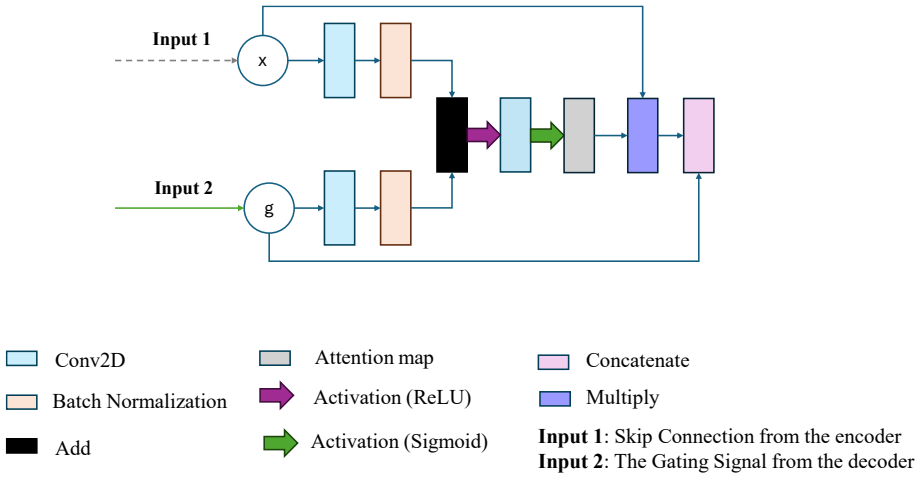

Fig. 3. Attention gate incorporated between skip connections and de-convs.

ResU-Net, another variant of baseline U-Net, incorporates residual information throughout its encoding and decoding blocks. The added residual mechanism has been shown in **Fig. 4.** In a standard U-Net, each block comprises of sequential convolutional blocks (Conv2D) as shown in **Fig. 2.,** but as the network gets deeper, it might lead to a "vanishing gradient" problem. To overcome this the Res U-Net model replaces the convolutional block with a 'Residual Block' as shown in **Fig. 4.** This block comprises of a main convolutional block along with a "identity shortcut" connection.

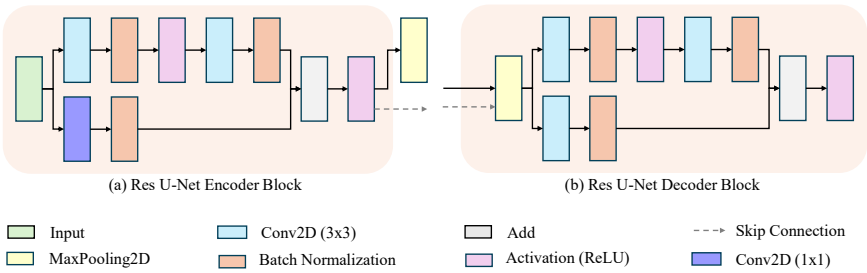

Fig. 4. (a) ResU-Net Encoder and (b) ResU-Net Decoder block.

U-Net++ extends the base U-Net architecture by using dense nested skip connections. The overall architecture has been shown in **Fig. 5.**, which is an inverse-pyramidal structure of convolutional nodes. Each convolution node is represented as $X^{i,j}$, in this notation i represents the down-sampling level along the encoder and the value of j represents the node's position along the diagonal dense skip path.

The architecture comprises three primary connections, the encoder path (down-sampling path) that connects nodes $X^{i,j}$ to $X^{i+1,j}$ via a MaxPool2D. The diagonal arrows (up-sampling path) that connect nodes $X^{i,j}$ to $X^{i-1,j+1}$ are Conv2DTranspose layers.

These help with up-sampling important feature maps from a deeper level to the next level. Finally, the Dense Skip Paths are the horizonal arrows that re-use the feature maps. In **Fig. 5.,** these arrows are represented with dotted arrows.

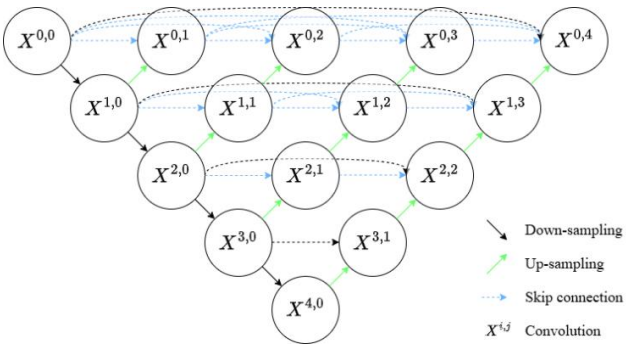

Fig. 5. U-Net++ architecture.

The foundational $X^{i,j}$, is detailed in **Fig. 6.** Each of these nodes process the input they receive from the connections discussed earlier.

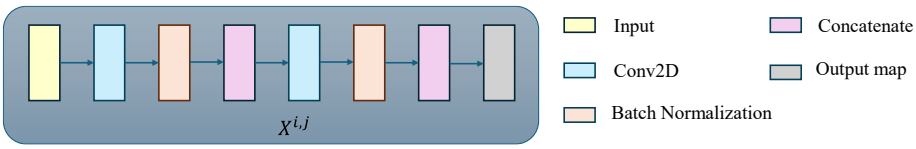

Fig. 6. Structure of an individual convolution node.

4 EXPERIMENTS AND RESULTS

4.1 Evaluation Metrics

Dice Coefficient.: Is the statistical measure of two sets which is used to determine the dice loss.

$$Dice\ Coefficient = \frac{2 \times X}{2 \times X + Y + FZ} : DC \in [0,1] \quad (1)$$

Where X represents True Positive, Y represents False Positive, and Z represents False Negative.

The Dice Coefficient value ranges from 0 to 1, where 0 represents no overlap and 1 represents perfect overlap between predicted and ground truth mask [11].

Jaccard Index.: It is the statistical measure to determine the Intersection over Union (IoU).

$$J(P,G) = \frac{|P \cap G|}{|P \cup G|} : J(P,G) \in [0,1] \tag{2}$$

Where *J (P, G)* is the Jaccard Index for the predicted mask *P* and the ground truth *G*. Like the Dice Coefficient, the Jaccard Index values also range between 0 and 1. A Jaccard index of 0 indicates no common elements whereas an index of 1 indicates that the two sets perfectly overlap [12].

4.2 Quantitative Analysis

The quantitative performance of the four architectures mentioned in **Section 3.3** on the test set is summarized in **Table 1**, reporting Dice Coefficient, Jaccard Index (IoU), and Test Loss. The models were tested on the same test set, this test set comprised 141 images, which were not used during the training process of these models.

Table 1. Performance Comparison of different U-Net Architecture.

Model	Dice Coefficient	Jaccard Index	Test Loss
U-Net	0.9560	0.9151	0.0445
ResU-Net	0.9627	0.9291	0.0378
Attention U-Net	0.9558	0.9149	0.0446
U-Net++	**0.9661**	**0.9343**	**0.0344**

As seen in **Table 1**, the U-Net++ had the best performance across all three metrics, achieving a Dice Coefficient of 0.9661, a Jaccard Index of 0.9343, and a Test Loss (Dice Loss) of 0.0344 [8]. The ResU-Net model performed similarly as the second-best performer across all three metrics [6]. On the other hand, the Baseline U-Net model and Attention U-Net model showed comparable performances, with Attention U-Net having a marginally lower performance compared to the U-Net Model [5][7].

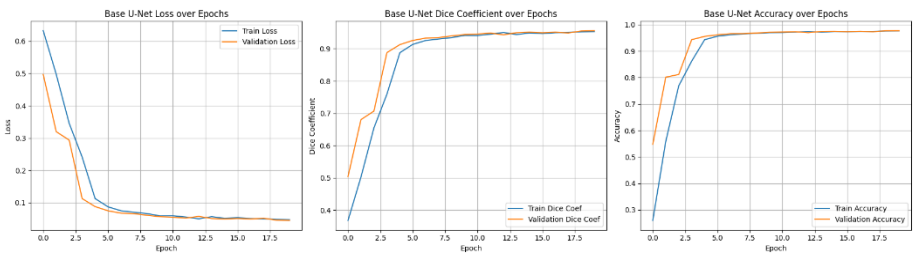

Fig. 7. Training and validation loss curves of the U-Net Model.

To understand the test results better, the training and validation loss and metric curves for the four models are illustrated in **Fig. 7. – Fig. 10.**

The U-Net model was used as a benchmark for all other models that were trained in the study. The U-Net model shows a steady increase in Training and Validation Accuracy, Dice coefficient, and Loss. The model converges around the 8th epoch and has a consistent but slight increase since then, indicating the model's stability.

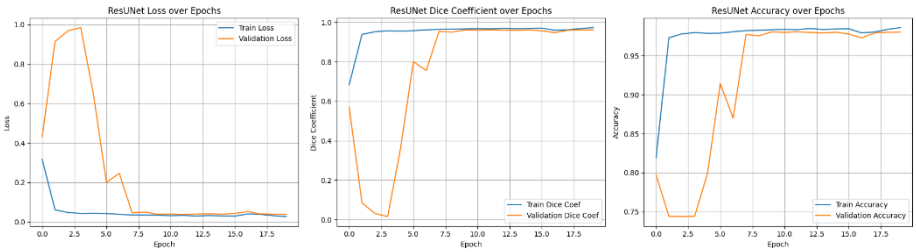

Fig. 8. Training and validation loss curves of the ResU-Net Model.

The ResU-Net model initially shows diverging graphs for Training and Validation in the first five epochs, but the model quickly stabilized, and the training and validation values converged around the 10th epoch, indicating the benefits of residual connections as discussed in **Section 3.3**.

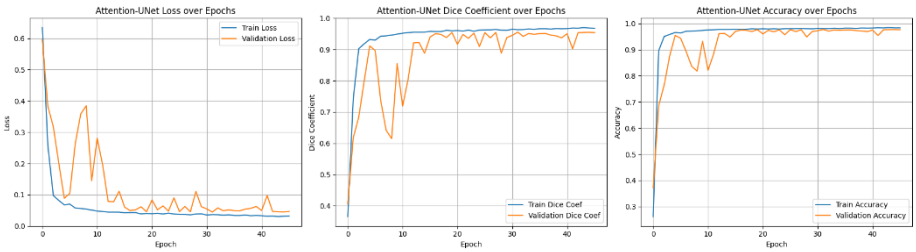

Fig. 9. Training and validation loss curves of the Attention U-Net Model.

The Attention U-Net model was trained with early stopping Reduce Learning Rate on Plateau callbacks due to the initial volatility in validation metrics, which can be observed in the graph. The model later steadily converged with an increase in validation and training metrics with the loss stabilizing around the 30th epoch. The model was set to be trained for 100 epochs, but due to the plateau and with no improvements from the Reduce Learning Rate on Plateau, the early stopping callback stopped the model training during the 44th epoch.

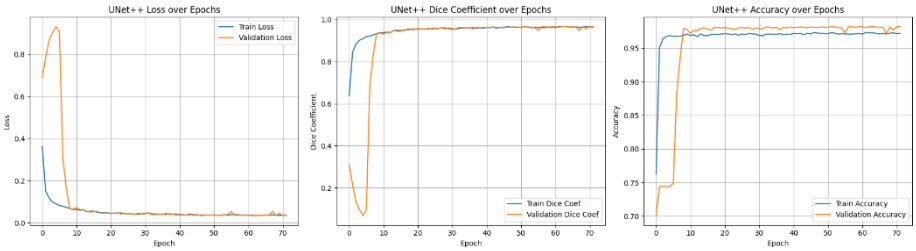

Fig. 10. Training and validation loss curves of the U-Net++ Model.

U-Net++ was trained with a 100-epoch limit, to mainly accommodate the complex structure of U-Net++. As seen from the graph, the model initially struggled to learn the data, which can be inferred from the increase in Dice Loss. With more epochs, the model slowly learned from the data and started to converge from the 5th epoch. After evaluating model training history on the 70th epoch, it was seen that the model did not display any volatility, and the validation and training metrics had converged and plateaued; hence the call to stop the training was taken on the 72nd epoch. This was done due to the long training time and limited T4 GPU resources on Google Colab.

While the simpler U-Net and ResU-Net models converged efficiently within 20 epochs, the more complex Attention U-Net and U-Net++ models required more optimized training approaches for optimal results. The graphs reveal that all models exhibited stable training without any overfitting, confirming the robustness of the training procedure.

Furthermore, we have evaluated the models over metrics like Accuracy, Specificity, Sensitivity, Precision and F1-Score [26][27].

Table 2. Classification Report for the 4 U-Net Variants.

Model	Accuracy	Specificity	Sensitivity	Precision	F1-Score
U-Net	0.9768	0.9860	0.9524	0.9586	0.9555
Res U-Net	0.9807	0.9894	0.9578	0.9686	0.9631
Attention U-Net	0.9773	**0.9946**	0.9305	**0.9831**	0.9561
U-Net++	**0.9821**	0.9901	**0.9611**	0.9708	**0.9660**

As indicated in **Table 2**, U-Net++ displayed the best performance in terms of Accuracy, Sensitivity and F1-Score; however, Attention U-Net had better performance in terms of Specificity and Precision.

The reason that Attention U-Net has better Specificity and Precision outcomes are directly attributable to a fundamental difference in design of the two models. Attention U-Net is designed to weigh purity more heavily, and so in places where Attention U-Net has been observed to perform well, it achieved better precision, specificity, and where lower false positive were desirable. Higher Precision means there are fewer False

Positives. Attention U-Net employs attention gates (AGs) to selectively pay attention to relevant features and reliable confidence in its prediction without contamination from irrelevant background noise.

U-Net++ weighs quality of completeness (Thoroughness) more heavily, and this is also the reason it achieves greater Sensitivity (Recall), F1-score and Accuracy. Higher Sensitivity means that of the total actual object pixels, a larger portion of those pixels would be included from the predictions (lower false negatives). It also utilizes dense, nested skip connections to aggregate a great range of multi-scale features contributing to a more robust model for overall finding and aggregating the entire object, especially the small and/or faint portion of it.

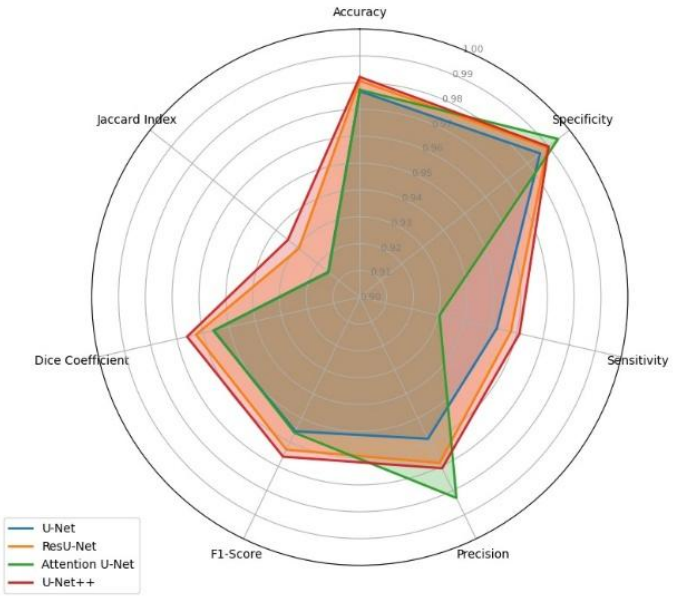

Fig. 11. Spider plot comparing segmentation performance of models based on different performance metrics.

Table 3. Boundary Distance Metrics for U-Net Variants.

Model	Hausdorff Distance (pixels)	ASSD (pixels)
U-Net	13.5736	1.7840
Res-UNet	12.4979	1.4583
Attention U-Net	13.4723	1.7355
U-Net++	**11.6183**	**1.3419**

The Hausdorff Distance (HD) [28] and Average Symmetric Surface Distance (ASSD) [29] are boundary distance metrics which assess the spatial precision and alignment of the segmentation boundary relative to the ground truth.

Table 3 summarized that all three modified U-Nets mentioned in **Section 3.3** exhibit improvements to boundary precision when benchmarked against the baseline U-Net. The baseline U-Net has the largest HD (13.5736) based upon the distance metric. Relatedly, it also has the highest ASSD (1.7840), suggesting a poor average alignment of boundaries in relation to the ground truth.

Alternatively, the U-Net++ displayed the best distance metric scores for both HD and ASSD (11.6183 and 1.3419 respectively). These boundary distance metrics denote that the modifications in the U-Net++ create the best conditions for precise spatial localization and statistically reliable reduction in error of segmentation. While it is true that Attention U-Net may have achieved higher pixel level purity (high Precision discussed above) the design did not translate superior boundary smoothness or alignment, compared to Res-UNet and UNet++ that exhibited improvement on both metrics.

4.3 Qualitative Analysis

To visually understand quantitative findings, this section presents the predicted segmentation from all four of the U-Net Architectures that were tested. **Fig. 12.** displays 6 sections: the initially provided X-ray image, the corresponding ground truth and predictions of the four models. The images used below are from the Test Set and have not been used to train the models.

Given the overall performance of all the models, as seen by their close quantitative scores (i.e., Dice Coefficient, within a range of 0.01) the visual distinction between them for a single image comparison might be negligible to the human eye. This high-level accuracy across all architectures underlines the efficiency of Deep Learning when it comes to segmentation tasks [2].

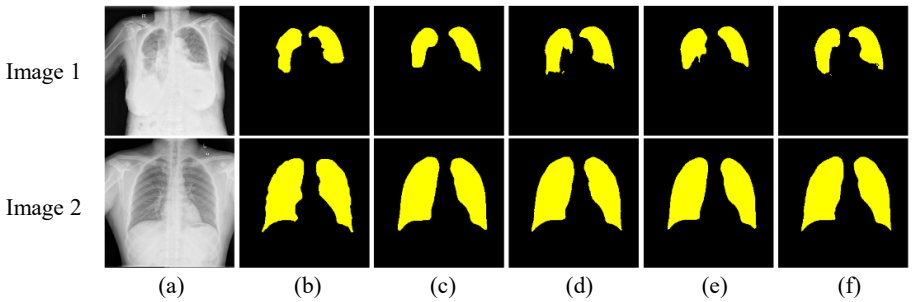

Fig. 12. Comparative qualitative results on test set examples.

The top row shows Example 1, and the bottom row shows Example 2. For example: (a) Original X-ray, (b) Ground Truth Mask, (c-f) Predicted lung masks for U-Net, ResU-Net, Attention U-Net, and U-Net++, respectively.

As observed in **Fig. 12.,** all models produce highly accurate lung segmentations that closely resemble the ground truth.

4.4 Statistical Analysis

This study conducted a statistical analysis to evaluate the differences in performance among the models mentioned in **Section 3.3**. The Friedman test [20][21] showed that there is a statistical difference in the mean Dice Coefficient [11] and Jaccard Index score [12] among all 4 models ($p < 0.001$).

Table 4. Statistical significance analysis using Nemenyi post-hoc test.

Model	U-Net	Attention U-Net	Res U-Net	U-Net++
U-Net	1.0000	0.9322	0.0000	0.0000
Attention U-Net	0.9322	1.0000	0.0000	0.0000
Res U-Net	0.0000	0.0000	1.0000	0.0011
U-Net++	0.0000	0.0000	0.0011	1.0000

Further, a post-hoc Nemenyi test [22][23][24] was conducted to examine the pair-wise difference. As shown in **Table 4**, U-Net++ emerged as the best model, showing statistical differences in its performance gain as compared to other models, which are U-Net ($p < 0.001$), ResU-Net ($p = 0.011$), and Attention U-Net ($p < 0.001$). ResU-Net was found to stand out second as it showed significant performance differences over both U-Net ($p < 0.001$) and Attention U-Net ($p < 0.001$). Performance amongst U-Net and Attention U-Net did not reveal any statistically significant differences ($p = 0.9322$), indicating their performance in the study was similar.

Concluding the statistical analysis, we can confirm that the differences in the model performances are significant, which establishes a clear ranking of model performances: **U-Net++** > ResU-Net > Attention U-Net ≈ U-Net.

5 DISCUSSION

This study evaluates four prominent U-Net architectures, as mentioned in **Section 3.3**, for automated lung segmentation of chest X-ray images. The results derived from this study showed significant insights into their performance and architectural strengths. In **Section 4.3** and **Section 4.4**, U-Net++ was established as the top-performing model, showing superior Dice Coefficient, Jaccard Index, Boundary Distances, lowest Test Loss and the most statistical difference from other models.

This robust performance of U-Net++ can be attributed to its densely connected skip pathways and multi-scale feature refinement, allowing it to handle the complexity of the lung structure. This architecture design suits the complex task of biomedical image segmentation in which accurate boundary mapping is crucial [13]. The model's dropout layers, and extended training periods also played a key role in ensuring optimal training

results. ResU-Net also demonstrated highly competitive results, having been trained for only 20 epochs, with the second-best performance metrics. The addition of residual blocks within the U-Net architecture contributed to better feature learning [14]. Overall, as seen from the qualitative analysis, all the models performed closely, demonstrating robustness of the U-Net architecture in lung segmentation [2].

Table 5. Evaluation of the four trained models in this study with the JSRT dataset [19].

Model	Dice Coefficient	Jaccard Index	Average Loss
U-Net	0.9783	0.9643	0.0356
ResU-Net	0.9793	0.9650	0.0348
Attention U-Net	0.9789	0.9646	0.0353
U-Net++	**0.9817**	**0.9699**	**0.0299**

The same four models as mentioned in **Section 3.3** were furthermore cross domain evaluated on a widely used dataset from the Japanese Society of Radiological Technology (JSRT) [19]. The dataset has 93 images with their corresponding masks. The results in **Table 5** correspond to the proposed comparison by our study as U-Net++ gives the best results in terms of both IoU value and Dice Coefficient (0.9699 and 0.9817 respectively). The rest of the results in **Table 5** also follow the same trend as proposed by the study.

A key strength of this study lies in the direct comparative analysis of these architectures on a combined dataset. As detailed in **Section 3.1**, this dataset consists of lung X-ray images from both healthy and tuberculosis-affected lungs [4]. This significantly enhances the models' ability to generalize and segment lung X-rays in various clinical scenarios, rather than being confined to only healthy anatomies as proven by **Table 5**. The model, having been trained in a real-world scenario, shows its robustness and demonstrates the models' potential ability to capture a wider range of anatomical variations apart from tuberculosis.

6 CONCLUSION

While this study works on a robust dataset with images of healthy and TB-affected lungs, its overall size could be considered moderate when compared to some large publicly available datasets for medical image segmentation [15]. While the models are effective, exploring ensemble methods by combining the strengths of different model architectures is crucial in refining and optimizing the model's performance [16]. Further improvements can be made by training the models on a set of more diverse X-rays having anatomical variations, tumors, viral and bacterial pneumonia, pulmonary edema and Chronic Obstructive Pulmonary Disease (COPD) [17]. This study benchmarks U-Net variants on the task of lung segmentation from X-Ray images, but lacks an extensive cross-domain evaluation to validate model transferability.

In conclusion, this study has successfully benchmarked the various U-Net models and analyzed their performance for automated lung segmentation in chest X-rays. The findings demonstrate that U-Net++ is the most effective architecture for this task among the models that were evaluated. The high performance of the models underlines the significant developments in deep learning for medical image analysis. These results will be helpful in integrating deep learning into a clinical environment, aiding in faster and more accurate diagnostic processes [18].

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